

Report on Jamming Simulation Environment and Execution of Basic Jamming in Pyjama Jamming Signals

Student ID: 1280391
Name: Natsuka Hosokawa

Kochi University of Technology

July 3, 2026

Table of Contents

- 1 Report on Server Environment Setup
- 2 Execution of Basic Jamming
 - Overview of Basic Jamming
 - Results of Basic Jamming
- 3 Conclusion and Future Work

Report on Server Environment Setup

Server Environment

- **Base OS:** Ubuntu 22.04 LTS (Stable Release)
- **Development Language:** Python 3.10 (OS Standard Package)
- **GPU Environment (TensorFlow 2.10 Compliant)**
 - CUDA Toolkit 11.2
 - cuDNN 8.1.1
- **Objective:** To safely and reliably reproduce the legacy GPU computing environment required by the Pyjama simulator within a modern container.

Important Note

This environment was forcefully adapted to the current server's GPU infrastructure. If migrating to a different environment in the future, there is a high probability that significant changes will be required.

Execution of Basic Jamming

Execution of Basic Jamming

1 Report on Server Environment Setup

2 Execution of Basic Jamming

- Overview of Basic Jamming
- Results of Basic Jamming

3 Conclusion and Future Work

Execution Overview of Basic Jamming

- **Jamming Type:** Basic Jamming (-3 dB jamming signal)
- **Execution Method:** Generation of jamming signals in an urban scenario on the Pyjama simulator.
- **Objective:** To quantitatively evaluate the impact of jamming signals on the Bit Error Rate (BER) of the communication.

Execution of Basic Jamming

1 Report on Server Environment Setup

2 Execution of Basic Jamming

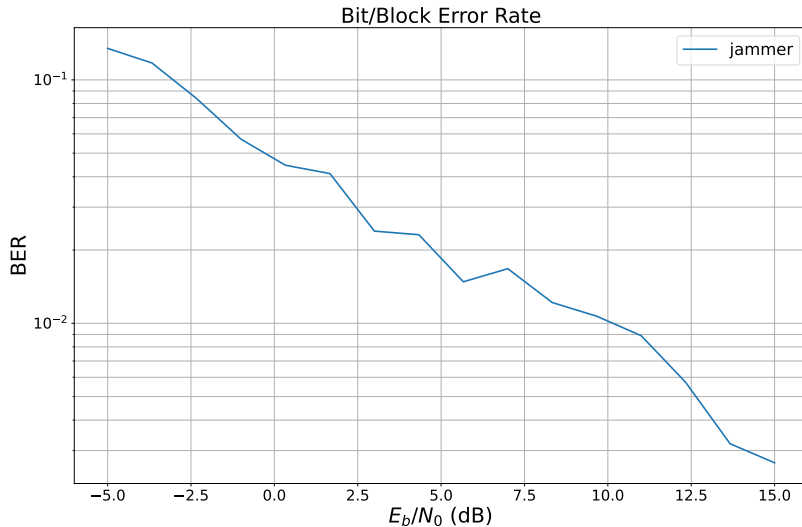
- Overview of Basic Jamming
- Results of Basic Jamming

3 Conclusion and Future Work

Measurement Results under Basic Jamming (−3 dB) Environment

EbNo [dB]	BER	BLER
−5.000	1.3472×10^{-1}	1.0000
−3.667	1.1727×10^{-1}	1.0000
−2.333	8.3861×10^{-2}	1.0000
−1.000	5.7059×10^{-2}	1.0000
0.333	4.4591×10^{-2}	1.0000
1.667	4.1223×10^{-2}	1.0000
3.000	2.3917×10^{-2}	9.9375×10^{-1}
4.333	2.3111×10^{-2}	9.1875×10^{-1}
5.667	1.4810×10^{-2}	7.6250×10^{-1}
7.000	1.6760×10^{-2}	7.4375×10^{-1}
8.333	1.2167×10^{-2}	6.2500×10^{-1}
9.667	1.0692×10^{-2}	4.6250×10^{-1}
11.000	8.8928×10^{-3}	3.4375×10^{-1}
12.333	5.7210×10^{-3}	2.8125×10^{-1}
13.667	3.1993×10^{-3}	2.3125×10^{-1}
15.000	2.6723×10^{-3}	1.2500×10^{-1}

BER Trend under Basic Jamming (-3 dB) Environment



Conclusion and Future Work

● Achievement of Environment Setup

- Successfully and completely reproduced the legacy GPU environment (e.g., TensorFlow 2.10) required to run the Pyjama simulator on a container.

● Confirmation of Basic Jamming (-3 dB) Impact

- Conducted baseline measurements using the POS mitigation system.
- Confirmed the theoretical behavior where the communication error rates (BER/BLER) decrease as the signal strength (E_b/N_0) increases, due to the relative decline in the jammer's influence (JSR).

● Future Work

- Implement "MASH", a more advanced multi-antenna mitigation algorithm, within this environment.
- Compare the simulation results between POS and MASH applications to evaluate the differences in mitigation performance.